

3.1 ADVANTAGE AGAINST ADVERSARIAL ATTACK

We attempt to explain the advantages of SAP under the assumption that we are applying it to a pre-trained model that achieves high generalization accuracy. For instance p under policy π , if the number of samples drawn for each layer i , r_p^i , is large, then fewer parameters of the neural network are pruned, and the scaling factor gets closer to 1. Under this scenario, the stochastically pruned model performs almost identically to the original model. The stochasticity is not advantageous in this case, but there is no loss in accuracy in the pruned model as compared to the original model.

On the other hand, with fewer samples in each layer, r_p^i , a large number of parameters of the neural network are pruned. Under this scenario, the SAP model’s accuracy will drop compared to the original model’s accuracy. But this model is stochastic and has more freedom to deceive the adversary. So the advantage of SAP comes if we can balance the number of samples drawn in a way that negligibly impacts accuracy but still confers robustness against adversarial attacks.

SAP is similar to the *dropout* technique due to Srivastava et al. (2014). However, there is a crucial difference: SAP is more likely to sample activations that are high in absolute value, whereas dropout samples each activation with the same probability. Because of this difference, SAP, unlike dropout, can be applied post-hoc to *pretrained models* without significantly decreasing the accuracy of the model. Experiments comparing SAP and dropout are included in section 4. Interestingly, dropout confers little advantage over the baseline. We suspect that the reason for this is that the dropout training procedure encourages all possible dropout masks to result in similar mappings.

3.2 ADVERSARIAL ATTACK ON SAP

If the adversary knows that our defense policy is to apply SAP, it might try to calculate the best strategy against the SAP model. Given the neural network h , input data x , corresponding true output y , a policy ρ over the allowed perturbations, and a policy π over the model parameters that come from SAP (this result holds true for any stochastic policy chosen over the model parameters), the adversary determines the optimal policy ρ^*

$$\rho^* = \arg \max_{\rho} \mathbb{E}_{p \sim \pi, r \sim \rho} [J(M_p(\theta), x + r, y)].$$

Therefore, using the result from section 2, the adversary determines the perturbation Δx as follows;

$$\Delta x = \arg \max_r r^\top \mathbb{E}_{p \sim \pi} [\mathcal{J}(M_p(\theta), x, y)].$$

To maximize the term, the adversary will set r to be in the direction of sign of $\mathbb{E}_{p \sim \pi} [\mathcal{J}(M_p(\theta), x, y)]$. Analytically computing $\mathbb{E}_{p \sim \pi} [\mathcal{J}(M_p(\theta), x, y)]$ is not feasible. However, the adversary can use Monte Carlo (MC) sampling to estimate the expectation as $\tilde{\mathcal{J}}(M_p(\theta), x, y)$. Then, using FGSM, $\Delta x = \lambda \cdot \text{sign}(\tilde{\mathcal{J}}(M_p(\theta), x, y))$.

4 EXPERIMENTS

Our experiments to evaluate SAP address two tasks: image classification and reinforcement learning. We apply the method to the *ReLU* activation maps at each layer of the pretrained neural networks. To create adversarial examples in our evaluation, we use FGSM, $\Delta x = \lambda \cdot \text{sign}(\mathcal{J}(M_p(\theta), x, y))$. For stochastic models, the adversary estimates $\mathcal{J}(M_p(\theta), x, y)$ using MC sampling unless otherwise mentioned. All perturbations are applied to the pixel values of images, which normally take values in the range 0-255. So the fraction of perturbation with respect to the data’s dynamic range would be $\frac{\lambda}{256}$. To ensure that all images are valid, even following perturbation, we clip the resulting pixel values so that they remain within the range $[0, 255]$. In all plots, we consider perturbations of the following magnitudes $\lambda = \{0, 1, 2, 4, 8, 16, 32, 64\}$.¹

To evaluate models in the image classification domain, we look at two aspects: the model accuracy for varying values of λ , and the calibration of the models (Guo et al., 2017). Calibration of a model is the relation between the confidence level of the model’s output and its accuracy. A linear calibration

¹All the implementations were coded in MXNet framework (Chen et al., 2015) and sample code is available at <https://github.com/Guneet-Dhillon/Stochastic-Activation-Pruning>

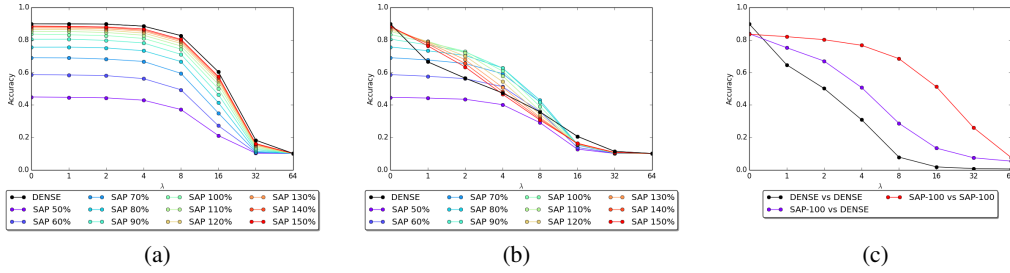


Figure 1: Accuracy plots of a variety of attacks against dense model and SAP models with different perturbation strengths, λ . For the SAP $\tau\%$ models, τ denotes the percentages of samples drawn from the multinomial distribution, at each layer. (a) SAP models tested against random perturbation. (b) SAP models tested against the FGSM attack, using MC sampling. (c) SAP-100 tested against an iterative adversarial attack, using MC sampling (legend shows defender vs. adversary). It is worth restating that obtaining the iterative attack of SAP models is much more expensive and noisier than the iterative attack of dense models.

is ideal, as it suggests that the accuracy of the model is proportional to the confidence level of its output. To evaluate models in the reinforcement learning domain, we look at the average score that each model achieves on the games played, for varying values of λ . The higher the score, the better is the model’s performance. Because the units of reward are arbitrary, we report results in terms of the the relative percent change in rewards. In both cases, the output of stochastic models are computed as an average over multiple forward passes.

4.1 ADVERSARIAL ATTACKS IN IMAGE CLASSIFICATION

The CIFAR-10 dataset (Krizhevsky & Hinton, 2009) was used for the image classification domain. We trained a ResNet-20 model (He et al., 2016) using SGD, with minibatches of size 512, momentum of 0.9, weight decay of 0.0001, and a learning rate of 0.5 for the first 100 epochs, then 0.05 for the next 30 epochs, and then 0.005 for the next 20 epochs. This achieved an accuracy of 89.8% with cross-entropy loss and *ReLU* non-linearity. For all the figures in this section, we refer to this model as the *dense* model. The accuracy of the dense model degrades quickly with λ . For $\lambda = 1$, the accuracy drops down to 66.3%, and for $\lambda = 2$ it is 56.4%. These are small (hardly perceptible) perturbations in the input images, but the dense model’s accuracy decreases significantly.

4.1.1 STOCHASTIC ACTIVATION PRUNING (SAP)

We apply SAP to the dense model. For each activation map $h^i \in \mathbb{R}^{a^i}$, we pick $k\%$ of a^i activations to keep. Since activations are sampled with replacement, k can be more than 100. We will refer to k as the percentage of samples drawn. Fig. 1a plots performance of SAP models against examples perturbed with random noise. Perturbations of size $\lambda = 64$ are readily perceptible and push all models under consideration to near-random outputs, so we focus our attention on smaller values of λ . Fig. 1b plots performance of these models against adversarial examples. With many samples drawn, SAP converges to the dense model. With few samples drawn, accuracy diminishes for $\lambda = 0$, but is higher for $\lambda \neq 0$. The plot explains this balance well. We achieve the best performance with $\sim 100\%$ samples picked. We will now only look at SAP 100% (SAP-100). Against adversarial examples, with $\lambda = 1, 2$ and 4 , we observe a 12.2%, 16.3% and 12.8% absolute increase in accuracy respectively. However, for $\lambda = 0$, we observe a 6.5% absolute decrease in accuracy. For $\lambda = 16$ again, there is a 5.2% absolute decrease in accuracy.

4.1.2 DROPOUT (DRO)

Dropout, a technique due to Srivastava et al. (2014), was also tested to compare with SAP. Similar to the SAP setting, this method was added to the *ReLU* activation maps of the dense model. We see that low dropout rate perform similar to the dense model for small λ values, but its accuracy starts decreasing very quickly for higher λ values (Fig. 2a). We also trained ResNet-20 models, similar

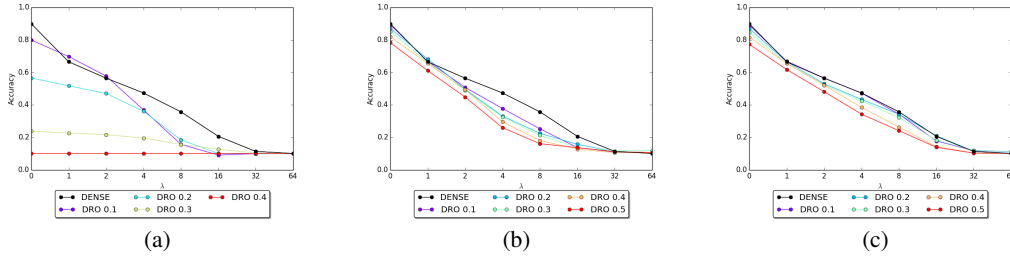


Figure 2: Robustness of dropout models, with different rates of dropout (denoted in the legends), against adversarial attacks, using MC sampling, with a variety of perturbation strengths, λ : (a) dropout is applied on the pre-trained models during the validations; (b) the models are trained using dropout, and dropout is applied during the validations; (c) the models are trained using dropout, but dropout is not applied during the validations.

to the dense model, but with different dropout rates. This time, the models were trained for 250 epochs, with an initial learning rate of 0.5, reduced by a factor of 0.1 after 100, 150, 190 and 220 epochs. These models were tested against adversarial examples with and without dropout during validation (Figs. 2b and 2c respectively). The models do similar to the dense model, but do not provide additional robustness.

4.1.3 ADVERSARIAL TRAINING (ADV)

Adversarial training (Goodfellow et al., 2014) has emerged a standard method for defending against adversarial examples. It has been adopted by Madry et al. (2017); Tramèr et al. (2017a) to maintain high accuracy levels even for large λ values. We trained a ResNet-20 model, similar to the dense model, with an initial learning rate of 0.5, which was halved every 10 epochs, for a total of 100 epochs. It was trained on a dataset consisting of 80% un-perturbed data and 20% adversarially perturbed data, generated on the model from the previous epoch, with $\lambda = 2$. This achieved an accuracy of 75.0% on the un-perturbed validation set. Note that the model capacity was not changed. When tested against adversarial examples, the accuracy dropped to 72.9%, 70.9% and 67.5% for $\lambda = 1, 2$ and 4 respectively. We ran SAP-100 on the ADV model (referred to as ADV+SAP-100). The accuracy in the no perturbation case was 74.1%. For adversarial examples, both models act similar to each other for small values of λ . But for $\lambda = 16$ and 32, ADV+SAP-100 gets a higher accuracy than ADV by an absolute increase of 7.8% and 7.9% respectively.

We compare the accuracy- λ plot for dense, SAP-100, ADV and ADV+SAP-100 models. This is illustrated in Fig. 3 For smaller values of λ , SAP-100 achieves high accuracy. As λ gets larger, ADV+SAP-100 performs better than all the other models. We also compare the calibration plots for these models, in Fig. 4. The dense model is not linear for any $\lambda \neq 0$. The other models are well calibrated (close to linear), and behave similar to each other for $\lambda \leq 4$. For higher values of λ , we see that ADV+SAP-100 is the closest to a linearly calibrated model.

4.2 ADVERSARIAL ATTACKS IN DEEP REINFORCEMENT LEARNING (RL)

Previously, (Behzadan & Munir, 2017; Huang et al., 2017; Kos & Song, 2017) have shown that the reinforcement learning agents can also be easily manipulated by adversarial examples. The RL agent learns the long term value $Q(a, s)$ of each state-action pair (s, a) through interaction with an environment, where given a state s , the optimal action is $\arg \max_a Q(a, s)$. A regression based algorithm, Deep Q-Network (DQN)(Mnih et al., 2015) and an improved variant, Double DQN (DDQN) have been proposed for the popular Atari games (Bellemare et al., 2013) as benchmarks. We deploy DDQN algorithm and train an RL agent in variety of different Atari game settings.

Similar to the image classification experiments, we tested SAP on a pretrained model (the model is described in the Appendix section A), by applying the method on the *ReLU* activation maps. SAP-100 was used for these experiments. Table 1 specifies the relative percentage increase in rewards of SAP-100 as compared to the original model. For all the games, we observe a drop in performance for